

VU DIEU MINH

(+84) 982343173 | vudieuminh1973@gmail.com

 [vudieuminh](#) |  [minhdieuvu08](#)

OBJECTIVE

An AI Engineer with a robust mathematical foundation and hands-on expertise in Computer Vision, LLM integration, and Reinforcement Learning. Proven ability to benchmark state-of-the-art models and build practical, pipeline-driven AI applications. Eager to apply my technical skills to solve business problems and help develop new products.

EDUCATION

- **VNUHCM — University of Science** 2022–2026 (expected)
Bachelor Ho Chi Minh City, Vietnam
 - Faculty of Mathematics and Computer Science
 - GPA: 3.53/4

SKILLS

- **Programming Languages:** C/C++, C#, Python, Common Lisp
- **Mathematical & Statistical Tools:** Matlab, R, $\text{\LaTeX}$
- **AI/ML Frameworks:** TensorFlow, PyTorch, scikit-learn
- **Developer Tools & Environments:** Unix/Linux, Git/GitHub
- **English:** Upper-Intermediate (TOEIC Listening-Reading: 770 | TOEIC Speaking-Writing: 270)

WORK EXPERIENCE

- **AI Engineer Intern** March 2025–June 2025
Tools: [Python, AWS Bedrock, LangChain, Amazon OpenSearch, Prompt Engineering, Vector Databases] 
 - Developed a chatbot that provides personalized fashion advice using Retrieval-Augmented Generation (RAG).
 - Implemented AWS Bedrock foundation models with a vector database (Amazon OpenSearch) for context-aware responses.
 - Applied prompt engineering techniques to ensure stylist-consistent and grounded answers.
 - Built an interactive Q&A system enabling natural conversation between users and the bot.

PROJECTS

- **Instance Segmentation of Medical Equipment in Hospital Environments** July 2026 – Present
PyTorch, Cascade Mask R-CNN, HTC, YOLO, Vision Transformers [Graduation Thesis]
 - Engineered and processed the HAMP dataset, specifically designed for detecting and segmenting medical equipment within complex hospital environments.
 - Benchmarked state-of-the-art architectures (**Cascade Mask R-CNN**, **HTC**, **Mask R-CNN**, **YOLOv8/v11**) with modern backbones (e.g., **ConvNeXt**, **PVT**) across 30+ configurations to evaluate the accuracy-efficiency tradeoff.
 - Achieved peak instance segmentation performance (mAP 57.4) using Hybrid Task Cascade with ConvNeXtv2-L, while analyzing FLOPs and FPS metrics for potential real-world healthcare deployment.
- **LLM-Guided Reinforcement Learning Agent** December 2025
Python, Gymnasium, Gemini API, Prompt Engineering 
 - Engineered a robust pipeline integrating **Google Gemini 2.5 API** to dynamically generate programmatic heuristics (Reward Shaping), jump-starting the RL agent's learning process.
 - Built a **Semantic Feature Extraction** module to process raw pixels into **structured JSON** (Scene Graphs), explicitly mitigating LLM spatial hallucinations and ensuring reliable API outputs.
 - Combined Tabular Q-Learning with a Population-Based Training framework, resulting in a significant increase in both average score and win rate over baseline models.
 - Designed custom data visualization pipelines using Matplotlib to effectively benchmark model performance and track heuristic accuracy.

• Mental Attention Classification Using EEG Data

December 2024

Programming Languages: Python



- Preprocessed raw **EEG signals** by extracting relevant channels and applying **normalization techniques** to ensure data consistency.
- Extracted **time-domain and frequency-domain features** (e.g., power spectral density, statistical metrics) to identify patterns across attention states.
- Trained and evaluated multiple classifiers (e.g., **Logistic Regression, SVM, Random Forest, Neural Networks**) using accuracy, precision, recall, and F1-score.
- Generated EEG signal plots, feature comparisons, **confusion matrices, and ROC curves** for model evaluation.

CERTIFICATIONS

- **Introduction to Game AI and Reinforcement Learning (Kaggle)** October 2025
- **Spring School on Statistics and Machine Learning (VIASM)** April 2025
- **Machine Learning Specialization by Andrew Ng (Coursera)** February 2025
- **Summer School on Mathematics (VIASM)** July 2024
- **Introduction to Machine Learning by Duke University (Coursera)** August 2023

LEADERSHIP & ACTIVITIES

- **Panelist | Women in Engineering and IT Workshop** [The IO Scholar](#) March 2026
 - Invited to share personal and academic experiences in Mathematical Methods in Informatics.
 - Inspired female students by discussing career orientation, research in Computer Vision/LLMs, and overcoming gender-related challenges in tech.
 - Panel discussion on integrating rigorous Math into modern AI development.